Project Report

Title: FSM-Based VGA Controller Design and Implementation Using SystemVerilog

Designed by:
Muhammad Muzammil Ahmed
Electronic Engineer

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VGA Controller

1) Introduction:

Video Graphics Array (VGA) is a widely used display standard that relies on precise timing signals to control pixel scanning and image rendering on a display. Generating these signals requires accurate coordination of horizontal and vertical synchronization pulses along with correct pixel positioning during the active display region.

This project focuses on the design and implementation of a VGA Controller using SystemVerilog, targeting a resolution of 640×480 pixels at a 60 Hz refresh rate. The controller is designed using a modular architecture, where each functional block is implemented as a separate module to improve readability, reusability, and verification.

A key aspect of this design is the use of a general-purpose finite state machine (FSM) to manage VGA timing. The same FSM-based Sync Pulse Generator is instantiated separately for horizontal and vertical synchronization, allowing consistent handling of VGA timing regions such as Sync Pulse, Back Porch, Active Display, and Front Porch.

The system also includes modules for pixel coordinate generation, frame buffering, and clock division. Verification of the design is performed through simulation using a layered testbench approach. Individual modules are also verified in isolation, followed by top-level integration testing to ensure correct interaction between blocks. RTL schematics and simulation waveforms are analyzed to validate functional correctness. The final design generates VGA-compliant synchronization signals and RGB output, making it suitable for FPGA-based display applications and further hardware deployment.

2) Description of Overall Architecture:

The VGA Controller is designed using a modular and hierarchical architecture to generate synchronization signals and pixel data compliant with the 640×480 resolution @60 Hz VGA standard. The system operates using a 25 MHz pixel clock, derived from the FPGA's 50 MHz clock, and coordinates horizontal and vertical scanning through synchronized timing logic.

At the core of the design is the general-purpose FSM-based Sync Pulse Generator, configured separately for horizontal and vertical timing. This FSM manages the VGA timing regions: Sync Pulse, Back Porch, Active Display, and Front Porch and generates the required synchronization pulses and active-region indicators.

A Pixel Generator module uses these timing signals to compute the current pixel (x) and line (y) coordinates during the active display region. These coordinates are then supplied to a Frame Buffer module, which stores image data in ROM and outputs the corresponding RGB pixel value.

An RGB data path ensures that valid pixel data is displayed only during the active video area, while blanking periods output black pixels. The Top-Level module integrates all submodules, manages signal routing, and provides VGA-compatible outputs including hsync, vsync, RGB channels, blanking, and clock signals.

This modular approach improves clarity, reusability, and ease of verification, making the design scalable and suitable for FPGA implementation.

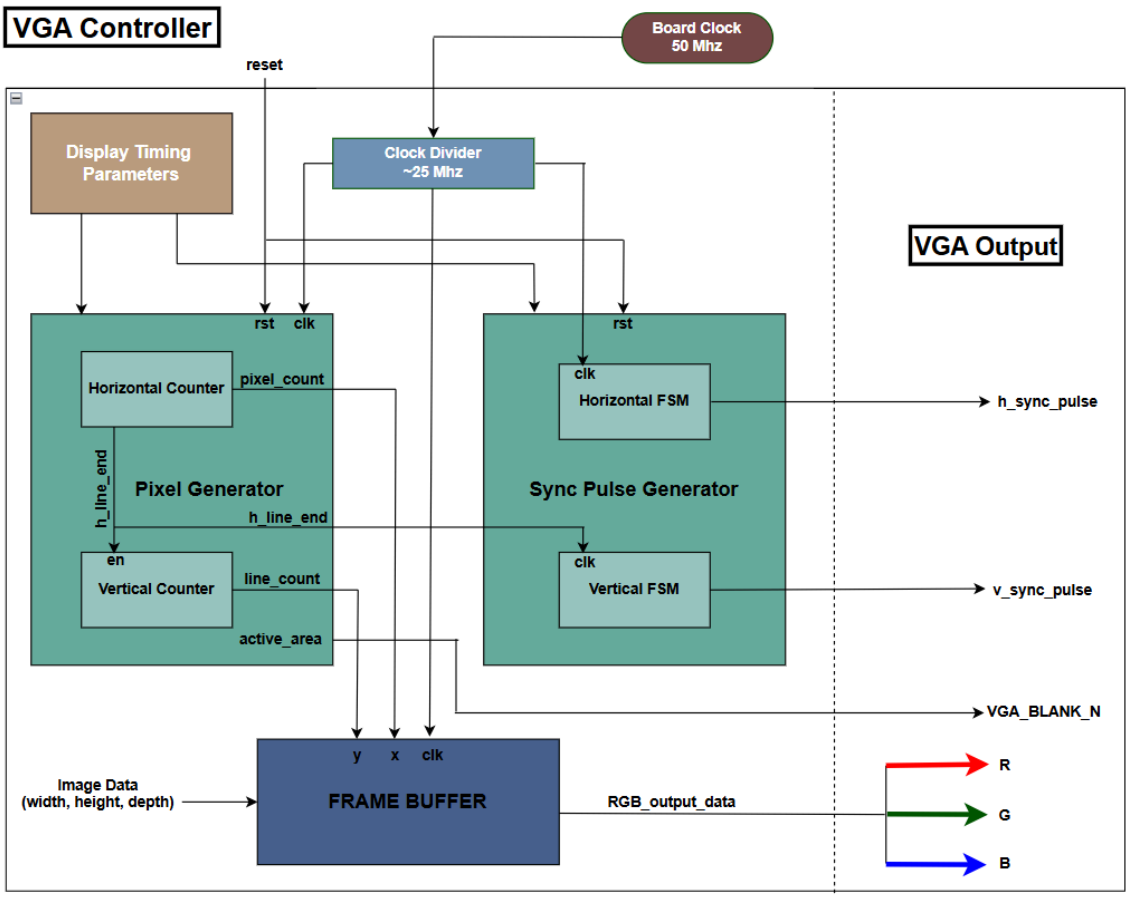


Figure: Architecture of VGA Controller

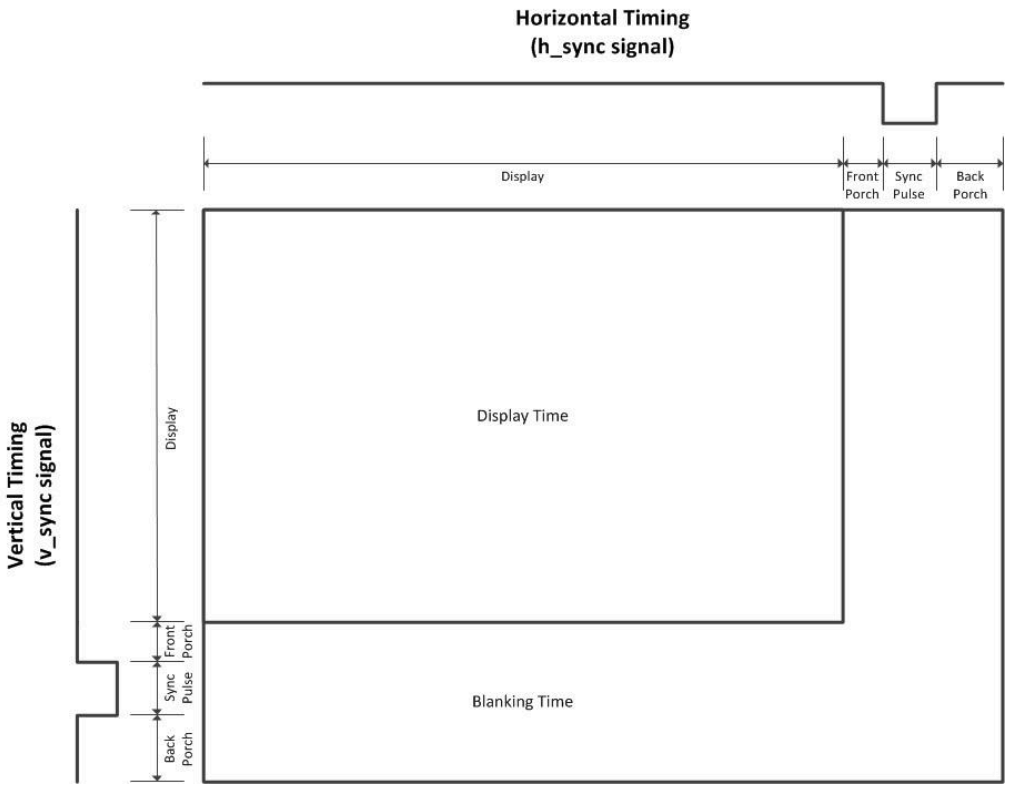


Figure: VGA Display

3) Design and Verification Tools:

The following software and hardware tools were used for the design and verification of the VGA controller:

- Intel Quartus Prime – RTL design and synthesis.
- QuestaSim – Simulation and functional verification.
- DE1-SoC FPGA Board – Hardware implementation and validation.
- SystemVerilog – Hardware description and verification language.

4) VGA Timing Specifications:

The standard timing for 640x480 @ 60 Hz is as follows:

Horizontal Timing:

- Active Region: 640 pixels
- Front Porch: 16 pixels
- Sync Pulse: 64 pixels
- Back Porch: 80 pixels
- Total: 800 pixels per line

Vertical Timing:

- Active Region: 480 lines
- Front Porch: 3 lines
- Sync Pulse: 4 lines
- Back Porch: 13 lines
- Total: 525 lines per frame

5) List of Modules:

- VGA Controller (Top-Level Module)
- Sync Pulse Generator Module
- Pixel Generator Module
- Frame Buffer Module
- Clock Divider Module

6) Module Description:

6.1) <u>VGA Controller Top Module:</u>

The Top Module integrates all functional blocks of the VGA controller and manages the overall system operation. It instantiates the clock divider, sync pulse generators, pixel generator, and frame buffer, and routes signals between them. This module produces the final VGA output signals including hsync, vsync, RGB color channels, blanking control, and pixel clock, representing the complete VGA controller system.

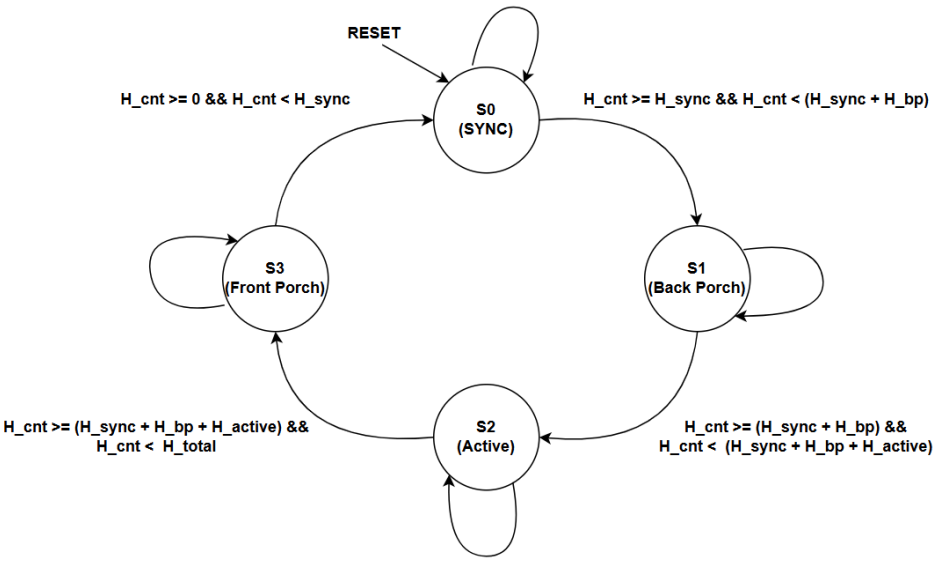
6.2) Sync Pulse Generator Module;

The Sync Pulse Generator implements a finite state machine (FSM) to generate VGA synchronization timing. Based on configurable timing parameters, it transitions through the following four standard VGA timing regions:

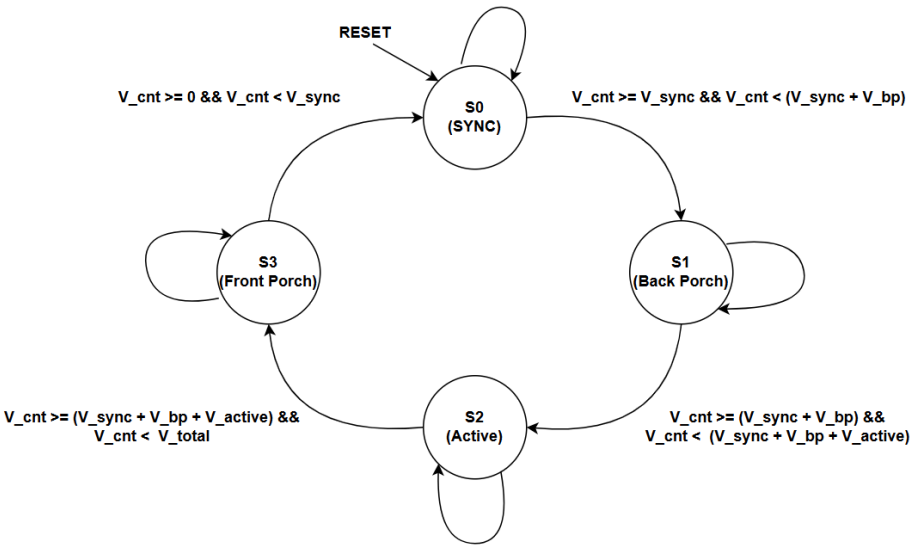
- Sync Pulse
- Front Porch
- Active Region
- Back Porch

The module generates an active-low synchronization pulse and an active-region indicator, and is reused for both horizontal and vertical timing to ensure modularity and consistency.

6.2.1) FSM for Horizontal Sync Pulse:



6.2.2) FSM for Vertical Sync Pulse:



6.2.3) FSM States Description:

State	State Description
S0 = Sync Pulse	Duration where sync signal is asserted
S1 = Back Porch	Idle time after sync pulse before active region
S2 = Active Region	Period where pixel data is valid and displayed
S3 = Front Porch	Short idle period after the active region

6.3) Pixel Generator:

The Pixel Generator tracks the current pixel and line positions by maintaining horizontal and vertical counters synchronized to the pixel clock. It determines whether the system is within the active display region and computes valid x and y pixel coordinates accordingly. These coordinates are used to correctly map image data to screen positions, while also generating an end-of-line signal for vertical synchronization.

6.4) Frame Buffer:

The Frame Buffer stores image data in a read-only memory structure and outputs pixel color values based on the current x and y coordinates. The module calculates a memory address from the pixel coordinates and retrieves the corresponding RGB data, enabling static image display on the VGA screen during the active display region.

6.5) Clock Divider Module:

The Clock Divider module generates a 25 MHz pixel clock by dividing the FPGA's 50 MHz input clock. This pixel clock drives all timing-sensitive VGA modules, ensuring compliance with VGA timing requirements for 640×480 resolution at 60 Hz.

7) Simulation Waveforms and Results:

To verify the functional correctness of the VGA controller prior to hardware implementation, layered testbench for the VGA controller was developed and comprehensive RTL simulations were performed using QuestaSim. Simulation waveforms and logs were analyzed to validate the behavior of the horizontal and vertical finite state machines, pixel coordinate generation, and synchronization signal timing. These simulation results provide confidence that the design operates correctly at the RTL level and aligns with the intended VGA standard before deployment on the FPGA.

7.1) VGA Controller Top Module:

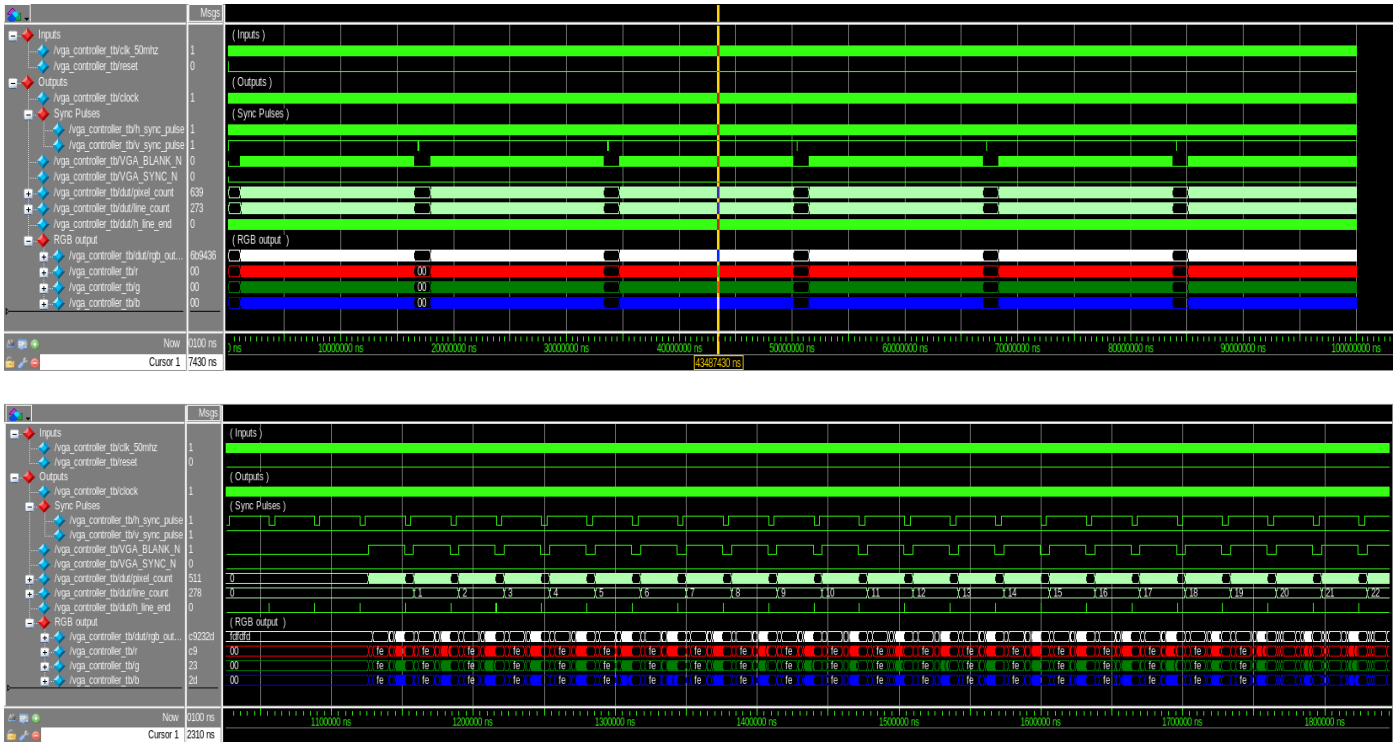


Figure: Simulation Waveform of RGB Output for Image Display

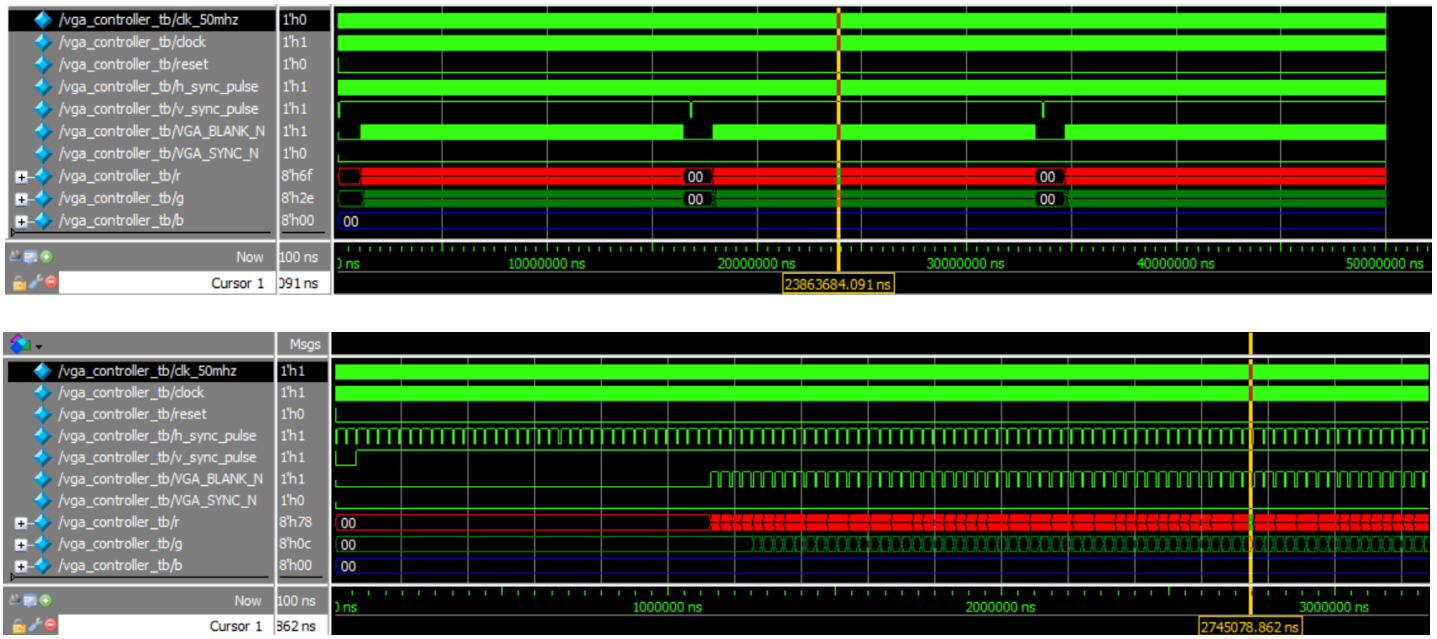


Figure: Simulation Waveform of RGB Output for Test Pattern

7.2) Sync Pulse Generator Module;

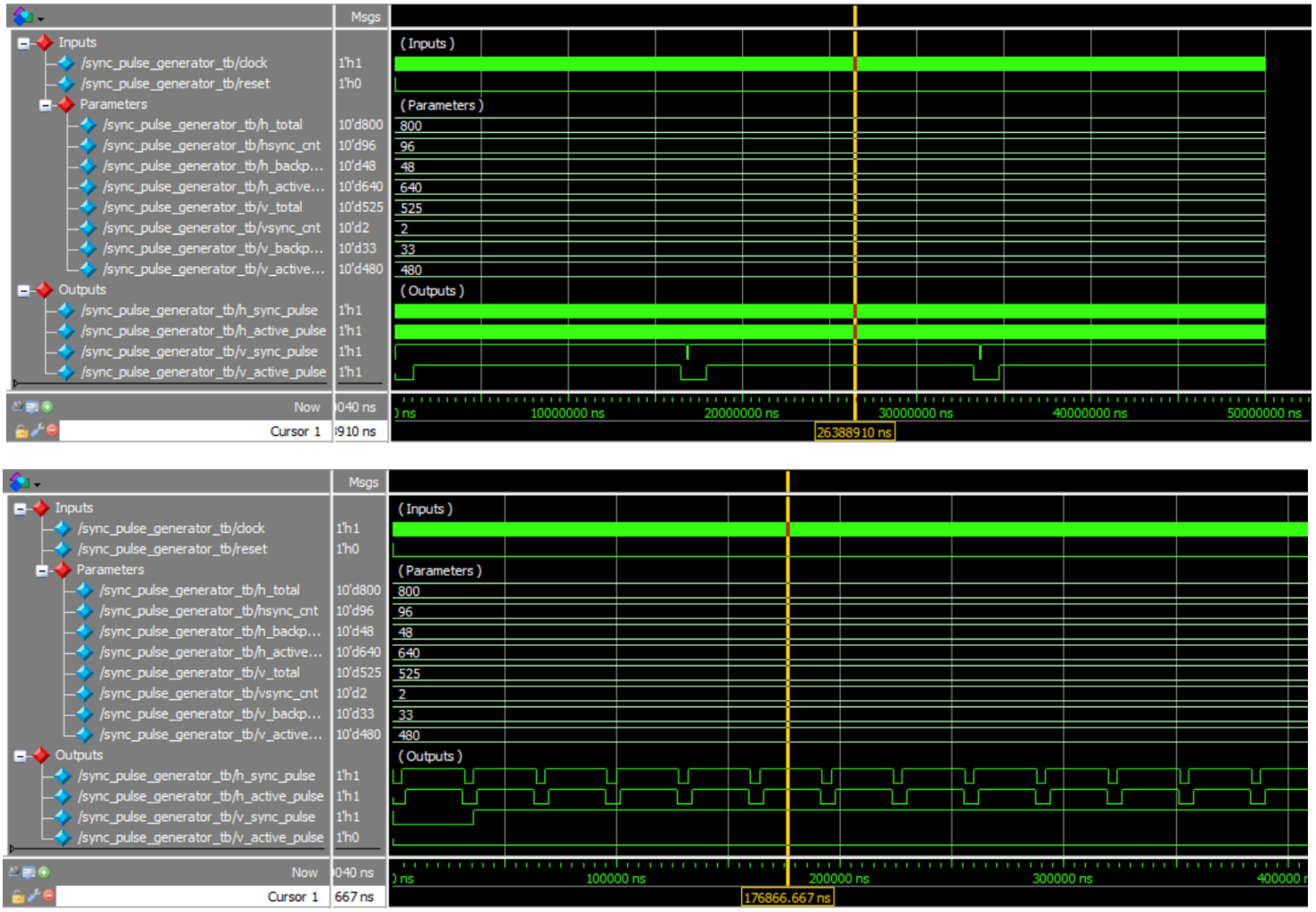


Figure: Simulation Waveform of VGA Synchronization Signals

7.3) Pixel Generator Module;

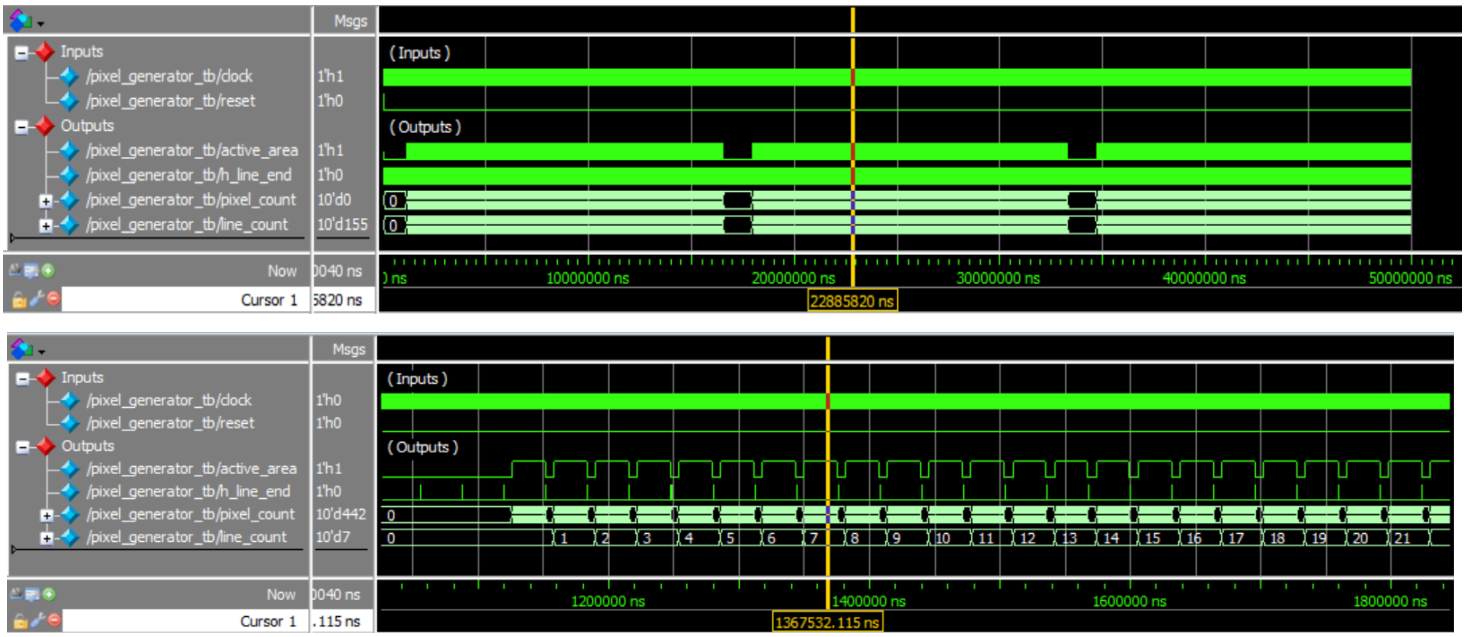


Figure: Simulation Waveform of Pixel Coordinates Generation

8) FPGA Implementation and Hardware Verification:

The following figures present the hardware implementation of the VGA Controller on the DE1-SoC FPGA board. The synthesized and compiled design was programmed onto the FPGA, with all necessary pin assignments configured in Quartus to map the FPGA I/O pins to the VGA connector signals, including hsync, vsync, and RGB channels. The VGA output was then verified by connecting the board to an LCD display through the VGA interface. Multiple test patterns, including solid colors, gradient and checker patterns, were evaluated to ensure proper pixel addressing, color generation, and synchronization. The correct generation of horizontal and vertical synchronization signals, along with the expected RGB output patterns, confirms the successful functionality of the controller on real hardware.9

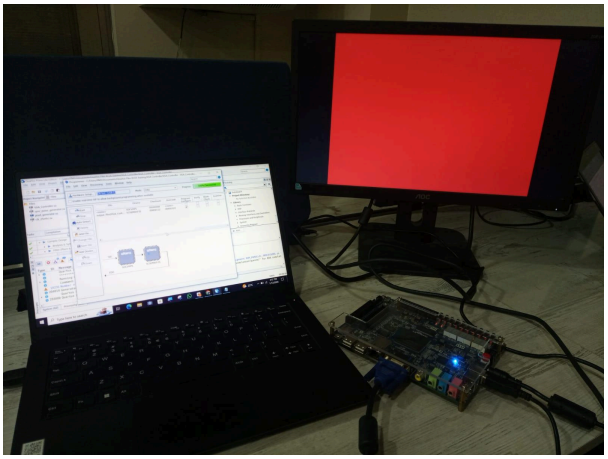


Figure: Test Pattern: Solid Color

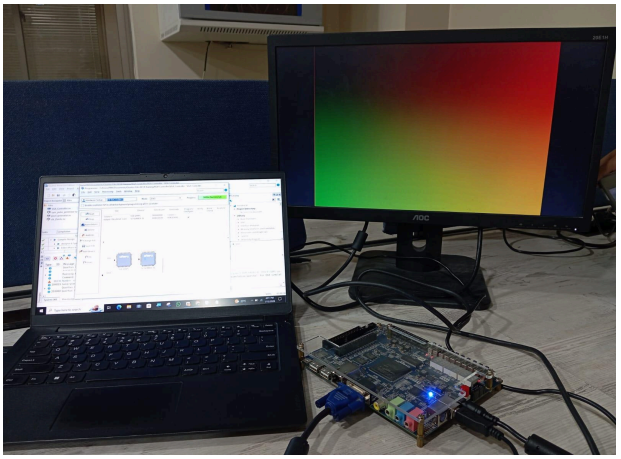


Figure: Test Pattern: Gradient Color

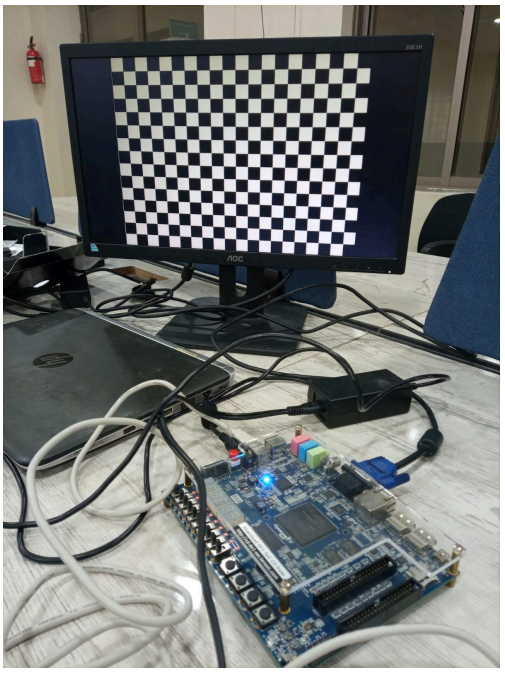
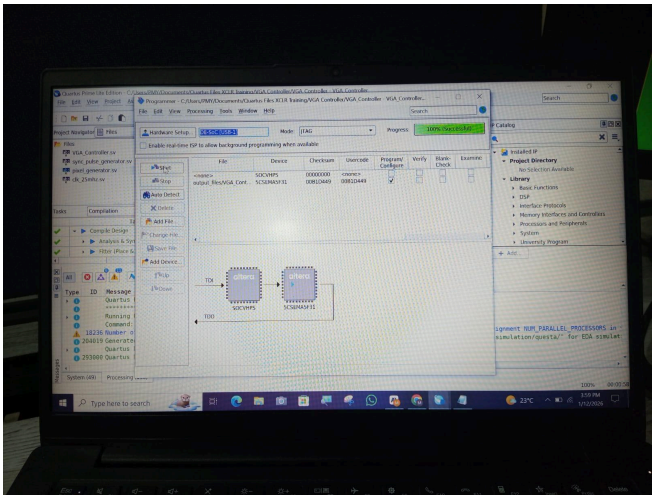


Figure: Test Pattern: Checkers



9) Conclusion:

In this project, a complete VGA Controller compliant with the 640×480 @ 60 Hz standard was successfully designed, implemented, and verified using a modular and FSM-based architecture. The design was developed in SystemVerilog, with separate finite state machines handling horizontal and vertical timing, enabling clean synchronization signal generation and accurate pixel addressing. Functional correctness was verified through RTL simulations using a layered testbench, ensuring that synchronization signals, pixel coordinates, and RGB outputs met the VGA timing requirements.

The validated design was then synthesized and implemented on the DE1-SoC FPGA, where proper pin assignments and on-board testing confirmed correct hardware operation. The successful display of multiple test patterns and image data demonstrates the reliability and robustness of the controller. Overall, this project strengthened practical understanding of digital video timing, FSM-based design, and FPGA-based hardware verification, and provides a solid foundation for extending the design to higher resolutions or more advanced video processing applications.